

NAME : SHAH FAISAL

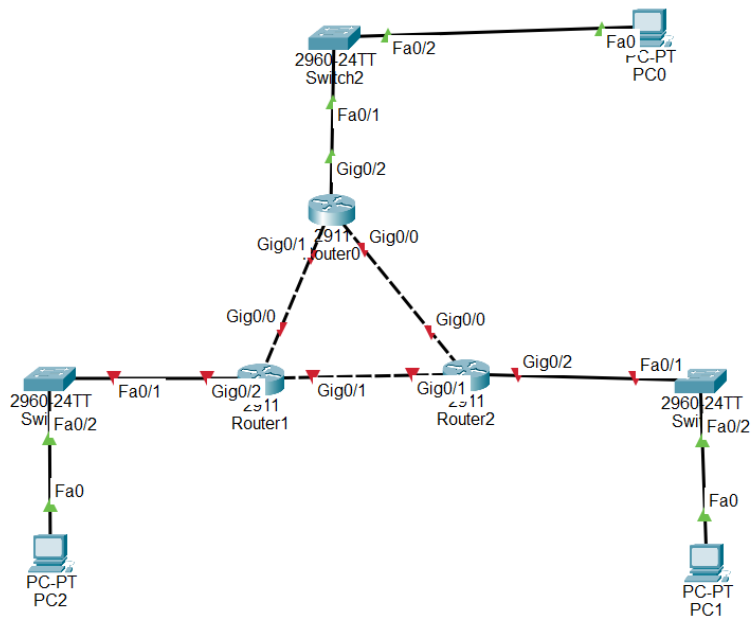
ROLL NO : 23I-0058

SECTION : BSE – 6A

Computer Networks
Lab 14

BSSE SPRING 2026

Lab 14



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Lab 14

Router0
— □ ×

```

CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 30.30.30.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed
state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed
state to up
          
```

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 PC-PT
PC2


 PC-PT
PC1

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed
state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/2, changed state to up

Router(config)#
```

PC-PT
PC2

PC-PT
PC1

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/2, changed state to up

Router(config)#
```

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PC2

PC-PT
PC1

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PC0

☐ DHCP ☒ Static

IPv4 Address: 192.168.30.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.30.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:F7FF:FE57:2E54

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

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PC2

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::200:CFF:FE21:D30A

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

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PC1
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☐ DHCP ☒ Static

IPv4 Address:

Subnet Mask:

Default Gateway:

DNS Server:

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address:

Link Local Address:

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

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 PC-PT
PC2


 PC-PT
PC1

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Router0
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```

GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

Router(config)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config)#router ospf 11
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 30.30.30.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
          
```

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PC2


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PC1

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Router1
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```

Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up

Router(config)#
Router(config)#configure terminal
      ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 11
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.10.0 0.0.0.255 area
0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 20.20.20.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
          
```

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PC2



PC-PT
PC1

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Router1
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Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 20.20.20.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#router ospf 11
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.10.0 0.0.0.255 area
0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 20.20.20.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
          
```

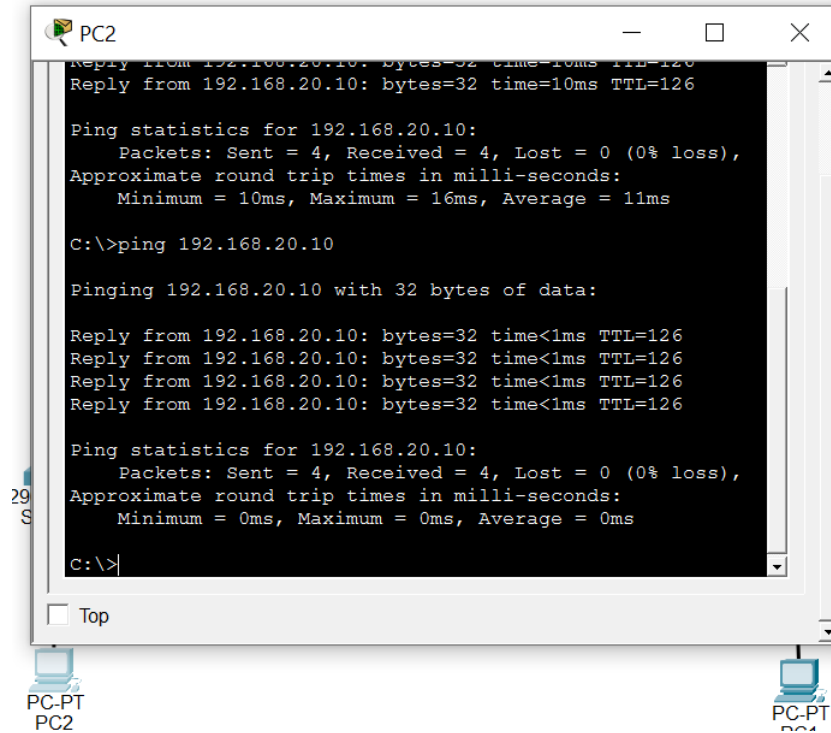


PC-PT
PC2



PC-PT
PC1

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```
PC2
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 16ms, Average = 11ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

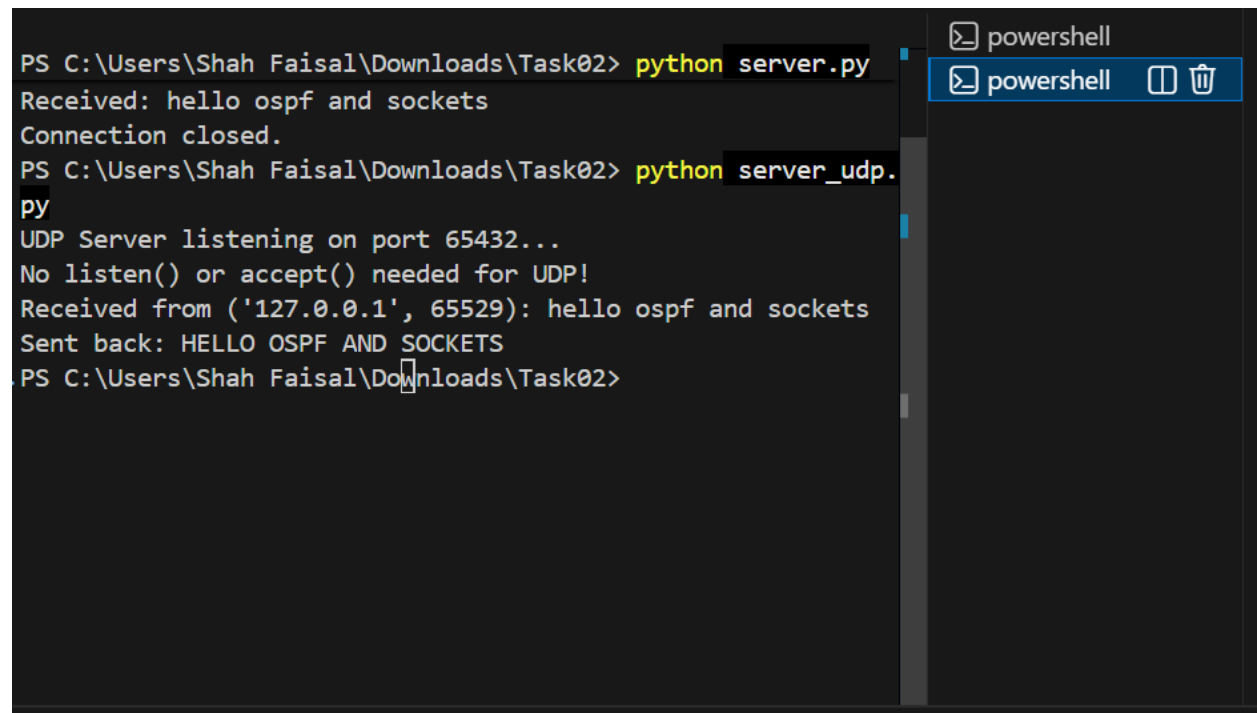
C:\>
```

PC-PT
PC2

PC-PT
PC4

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TASK 02



```
PS C:\Users\Shah Faisal\Downloads\Task02> python server.py
Received: hello ospf and sockets
Connection closed.
PS C:\Users\Shah Faisal\Downloads\Task02> python server_udp.
py
UDP Server listening on port 65432...
No listen() or accept() needed for UDP!
Received from ('127.0.0.1', 65529): hello ospf and sockets
Sent back: HELLO OSPF AND SOCKETS
PS C:\Users\Shah Faisal\Downloads\Task02>
```

powershell

powershell

```
PS C:\Users\Shah Faisal\Downloads\Task02> python client.py
Connected to server at 127.0.0.1:65432
Sending: hello ospf and sockets
Server replied: HELLO OSPF AND SOCKETS
PS C:\Users\Shah Faisal\Downloads\Task02> python client_udp.py
Sending (UDP): hello ospf and sockets
Datagram sent (no guarantee of delivery!)
Server replied (UDP): HELLO OSPF AND SOCKETS
PS C:\Users\Shah Faisal\Downloads\Task02>
```

powershell

powershell